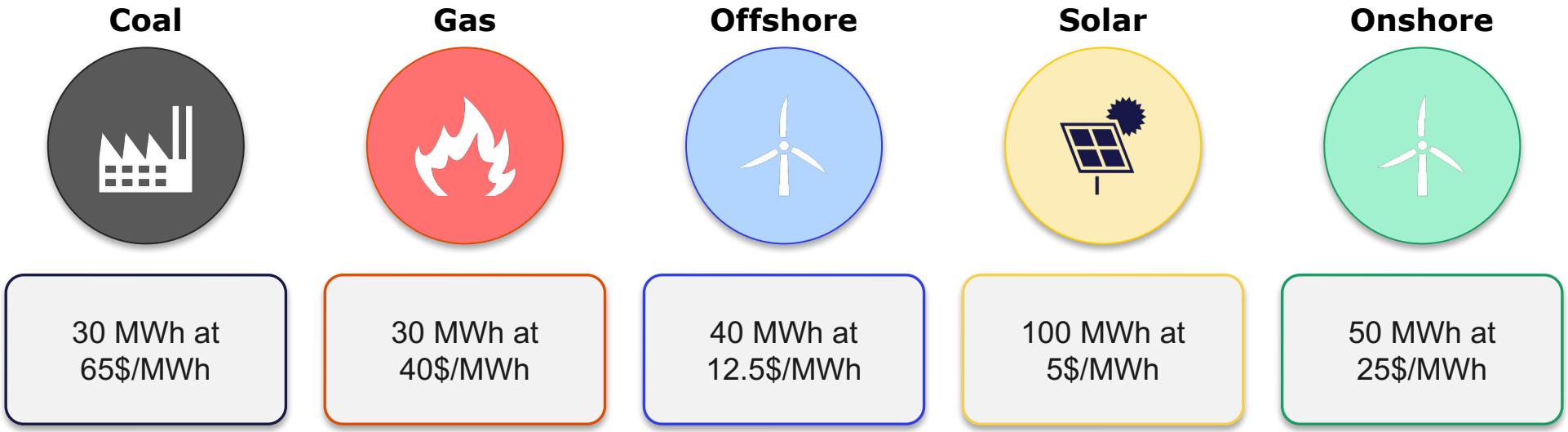
Game Theory in Electricity Markets

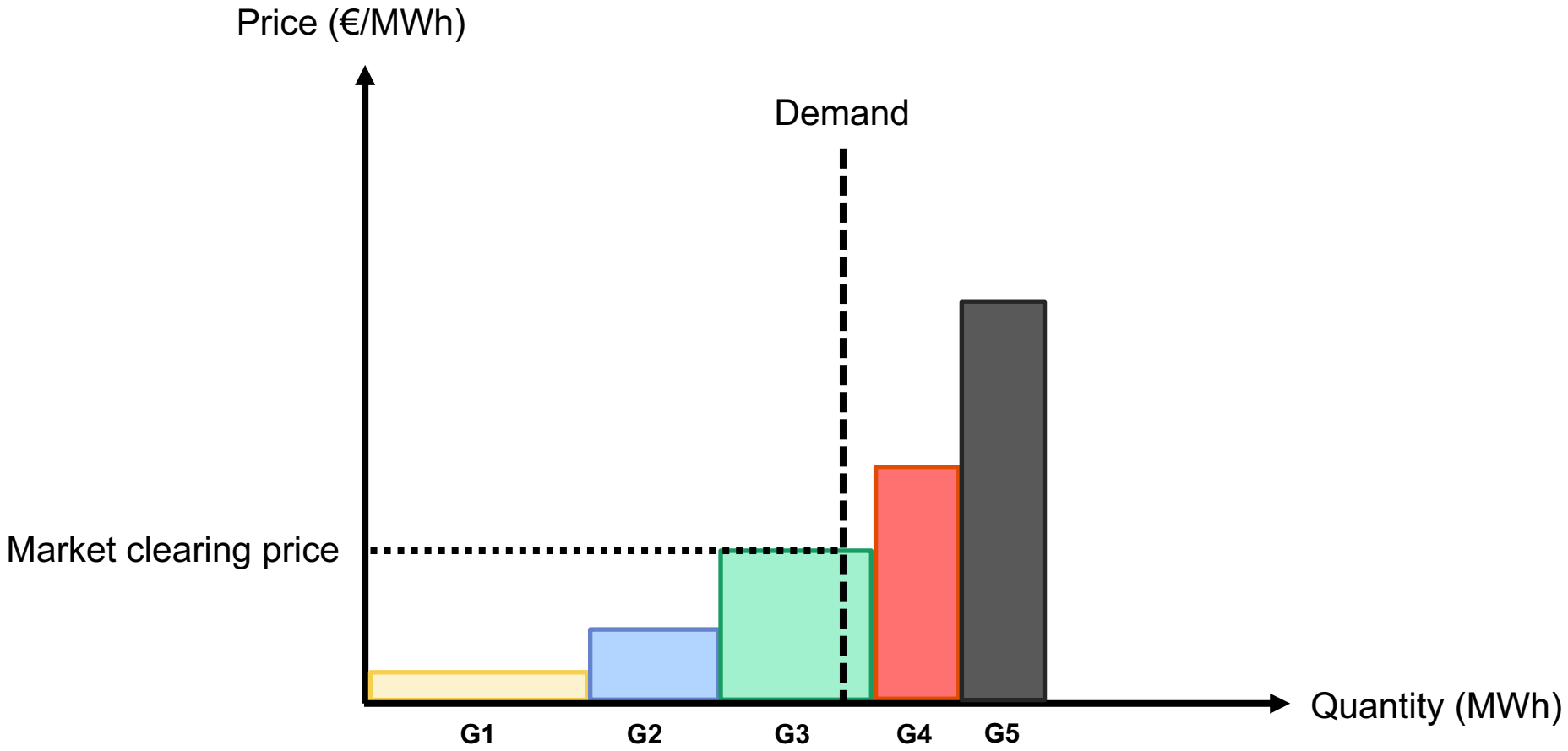
Week 5 – Modelling electricity markets

How to fulfill the demand in the best way?

We just ask every generator to tell us how much it costed them to produce.

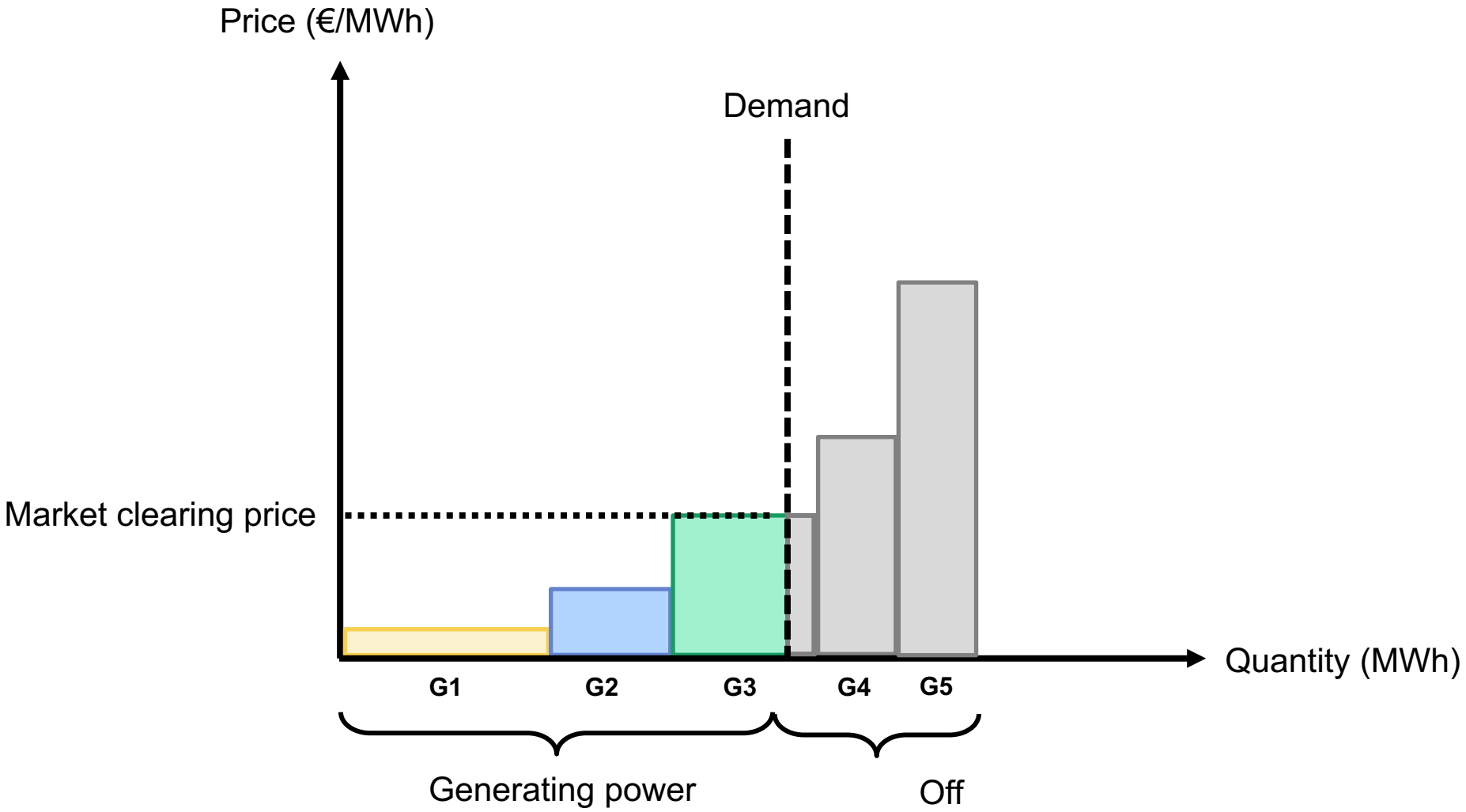


Market Clearing 101

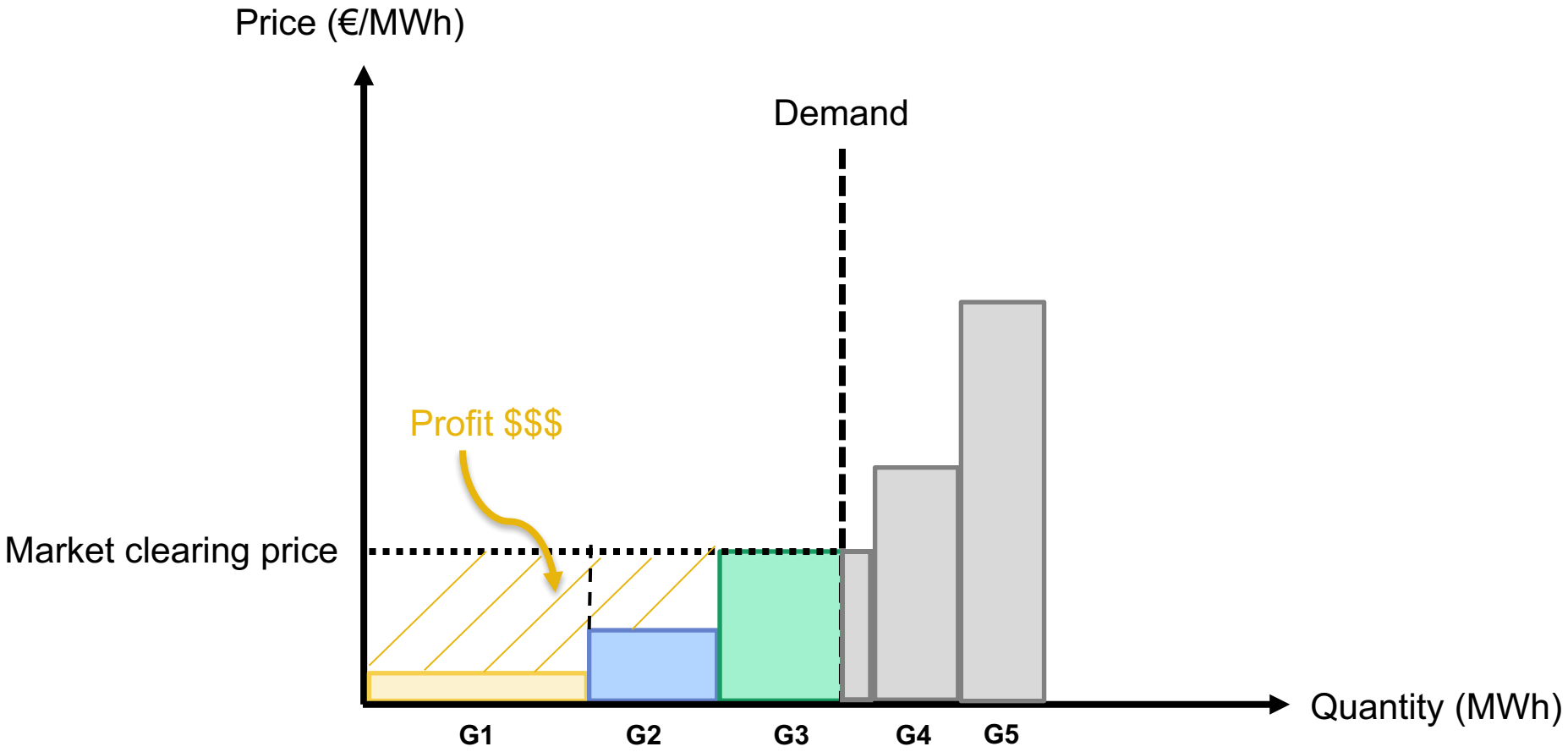


Graphic from: https://www.britannica.com/

Market Clearing 101



Market Clearing 101



But what really happened?

The operator solved a problem of the form

$\min_x \ \Sigma_i \ c_i(x_i)$

$s.t. \ \ \Sigma_i \ x_i = D$

$x \in \mathcal{X}$

- **Variables:** x_i = production of generator i, D = demand
- **Costs:** $c_i(x_i) = a_i x_i$ are linear, with a_i the marginal cost of generator i (€/MWh)
- **Feasible set:** $\mathcal{X}$ = { x : $0 \leq x_i \leq \bar{x}_i$ for all i }, i.e. each generator is limited by its capacity $\bar{x}_i$

Cost functions

In reality, every generator bids several blocks

- Each block has a size and a price, so the cost function is piecewise linear
- Block sizes and prices are the parameters a generator optimizes to maximize its revenue

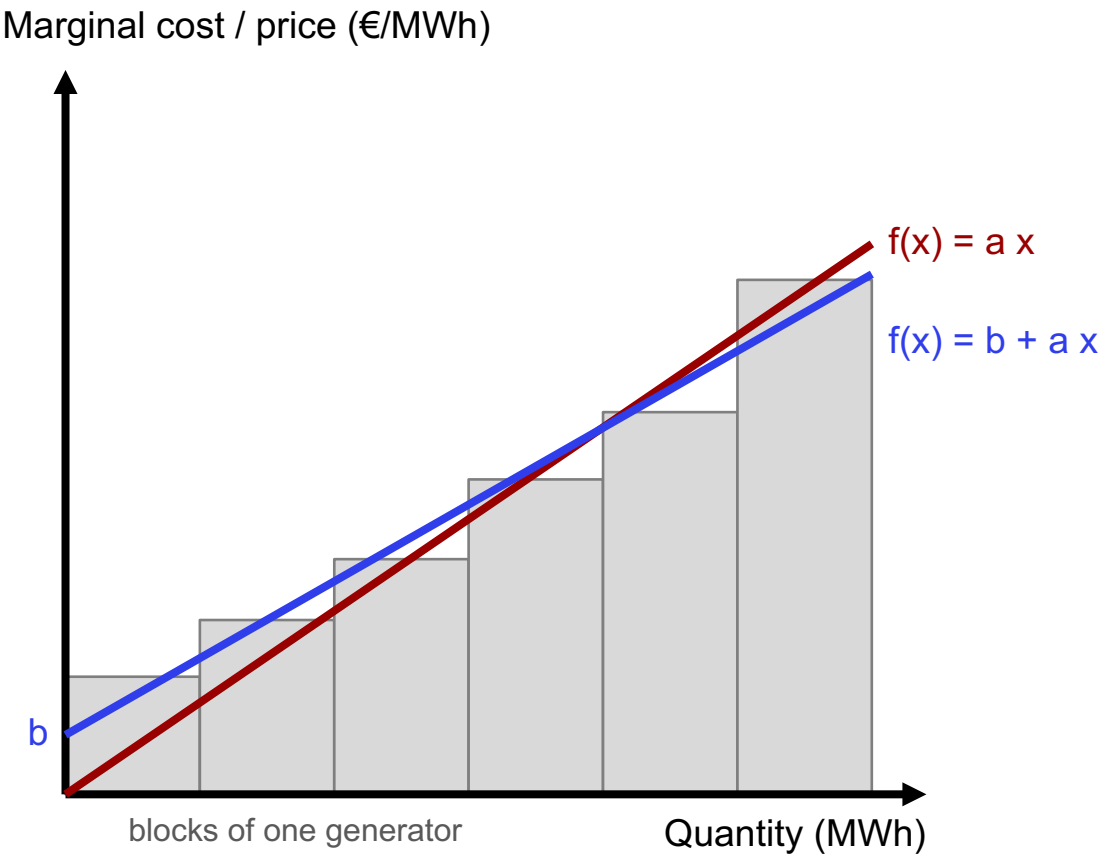
Or use a smooth approximation

Fully quadratic (line through the origin, f(0) = 0)

$$c_i(x) = \tfrac{1}{2}\, a_i\, x^2$$

Quadratic + linear (line with intercept)

$$c_i(x) = \tfrac{1}{2}\, a_i\, x^2 + b_i\, x$$



Capacity constraints

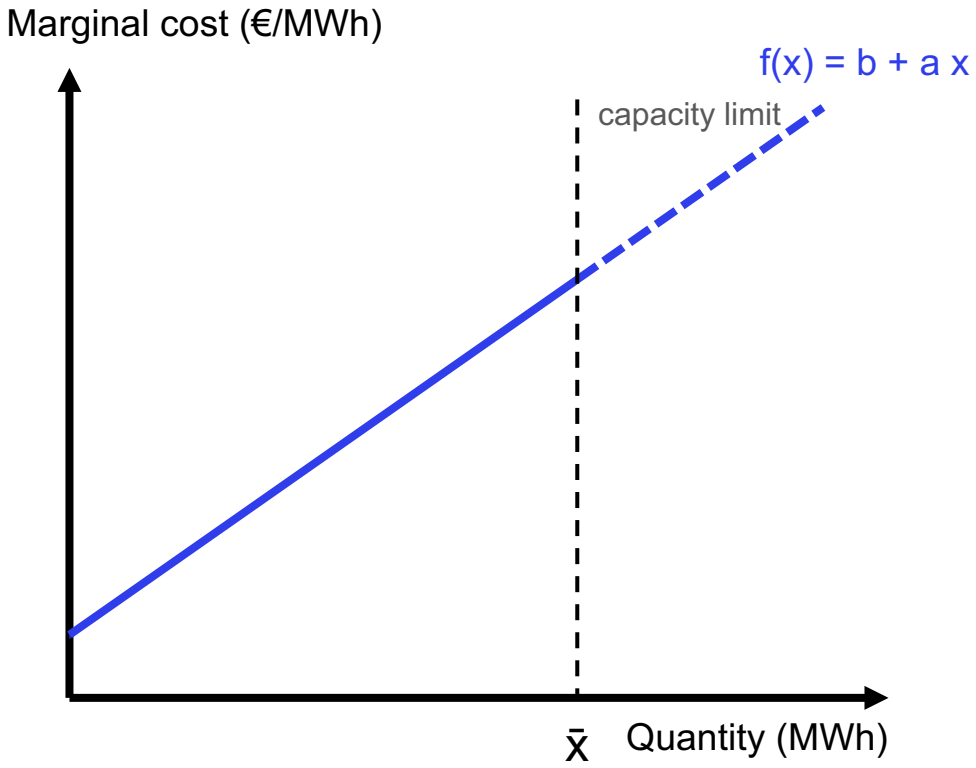
Each generator is limited by its capacity, which gives the constraints

$$0 \le x_i \le \bar{x}_i \quad \text{for all } i$$

- These are linear constraints (a lower and an upper bound on each x_i)
- Common: the same limit for all generators

$$0 \le x_i \le \bar{x} \quad \text{for all } i$$

- With the smooth cost functions the limit is often not needed: the cost becomes so high for large x_i that the capacity is only weakly enforced



Network constraints

For every line (i, j) between two buses i and j:

Phase angle constraint (power flow on the line)

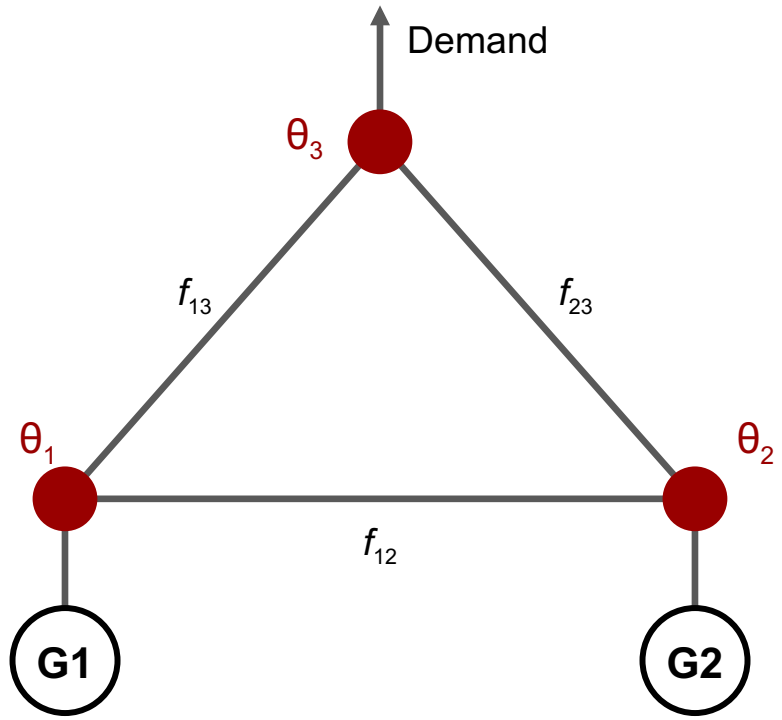
$$f_{ij} = b_{ij} \sin(\theta_i - \theta_j)$$

Maximum phase angle

$$|\theta_i - \theta_j| \leq \theta_max$$

- Technically not linear because of the sine. If the angles stay small, $\sin(\theta) \approx \theta$, so the constraint becomes linear:

$$f_{ij} = b_{ij} (\theta_i - \theta_j)$$



Intertemporal constraints

In reality, the market is cleared for the whole next day at the same time

The operator therefore solves the problem for all periods t = 1, …, T (e.g. 24 hours) at once:

$$\min_x \sum_t \sum_i c_{i,t}(x_{i,t})$$

$$\text{s.t. } \sum_i x_{i,t} = D_t \quad \text{for all } t$$

$$x_t \in \mathcal{X}_t \quad \text{for all } t$$

Intertemporal constraints couple consecutive periods. The most typical ones:

Ramp up / ramp down limits

$$-R_i \le x_{i,t+1} - x_{i,t} \le R_i$$

Linear: the difference of two variables is bounded by a constant (two linear inequalities)

Storage / reservoir level

$$s_{i,t+1} = s_{i,t} - x_{i,t}\ , \quad 0 \le s_{i,t} \le \bar{s}_i$$

Linear: the energy balance and the level limits are linear in x and s

Start-up, minimum up/down times

$$u_{i,t} \in \{0, 1\}$$

Not linear: the on/off status is binary, so the problem becomes mixed-integer and non-convex

Cost functions – which one to choose

- To consider the cost functions as piecewise linear is most realistic, although the quadratic approximations are also completely valid
- It gives players more variables to manipulate, therefore it may have highest PoA
- To approximate quadratically gives continuity of the derivative. This is needed for reformulating the operator's problem via its KKT conditions
- This also gives uniqueness of the dispatch. There are no draws as in the block bidding

⟶ It depends on what you want to do. Generally, when you work more theoretical it makes sense to go with one of the quadratic ones.

Fixed socially optimal profile revisited

Theorem (Strauch, Kamgarpour, Tsaousoglou, still not yet out). Fix A* as the set of socially optimal joint action profiles. Then the following programs yield an upper bound to the Price of Anarchy that is weakly tighter than the one of generalized smoothness.

$$\inf_{\lambda>0,\ \mu<1,\ a^* \in A^*} \quad \lambda / (1 - \mu)$$

$$\text{s.t.} \quad \Sigma_i\, c_i(a^*_i, a_{-i}) - \Sigma_i\, c_i(a) + C(a) \leq \lambda\, C(a^*) + \mu\, C(a) \quad \forall\ a \in A$$

change of variables →

$$\inf_{\rho>0,\ v>0,\ a^* \in A^*} \quad \rho$$

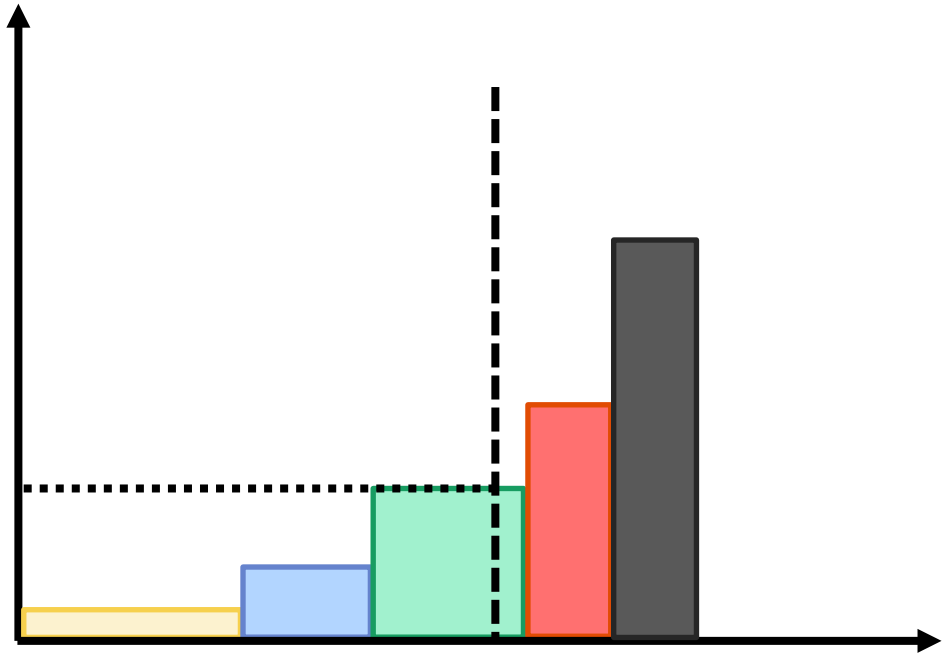
$$\text{s.t.} \quad C(a) + v\,[\,\Sigma_i\, c_i(a^*_i, a_{-i}) - \Sigma_i\, c_i(a)] \leq \rho\, C(a^*) \quad \forall\ a \in A$$

The problem: the set A* does generally not have any nice properties.

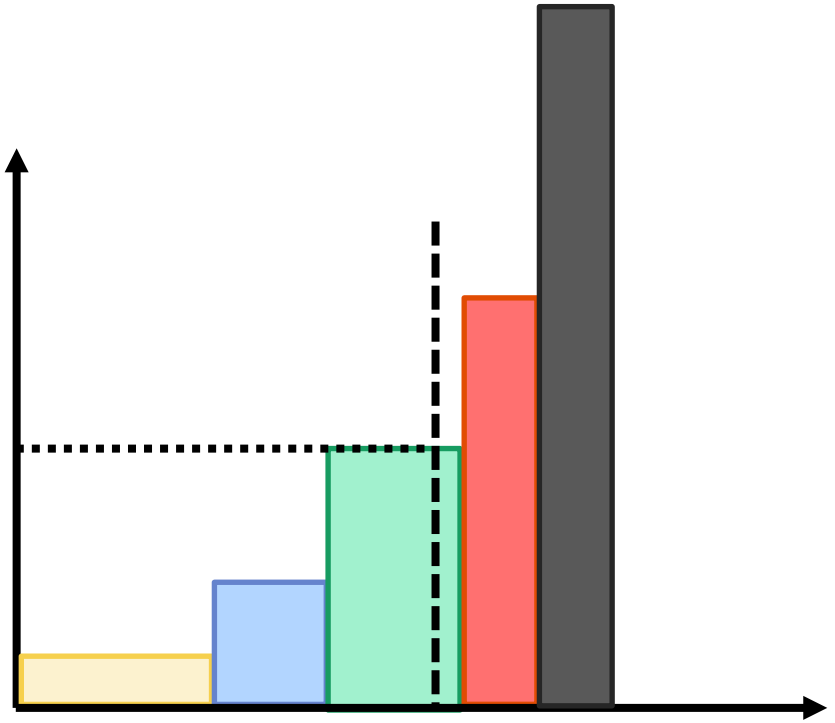
The solution: in electricity markets a relevant subset of A* can be parameterized

Market Clearing 101

Everybody declares their costs truthfully



Everybody bids double their price



Graphic from: https://www.britannica.com/

Get hands on A*

$$\min_x \sum_{t=1}^{T} \sum_{i=1}^{n} c_{i,t}(x_{i,t})$$

$$\text{s.t.} \sum_{i=1}^{n} x_{i,t} = D_t \quad \text{for all } t$$

If every generator tells us their true cost the resulting dispatch will be optimal

$$\min_x \sum_{t=1}^{T} \sum_{i=1}^{n} \omega c_{i,t}(x_{i,t})$$

$$\text{s.t.} \sum_{i=1}^{n} x_{i,t} = D_t \quad \text{for all } t$$

If they all give us their true cost multiplied by a factor, it is also socially optimal

The joint action ωc_t is in A* for all positive scalars ω

Our final problem (this time for real)

$$\inf_{\lambda>0,\ \mu<1,\ \omega>0} \quad \lambda / (1 - \mu)$$

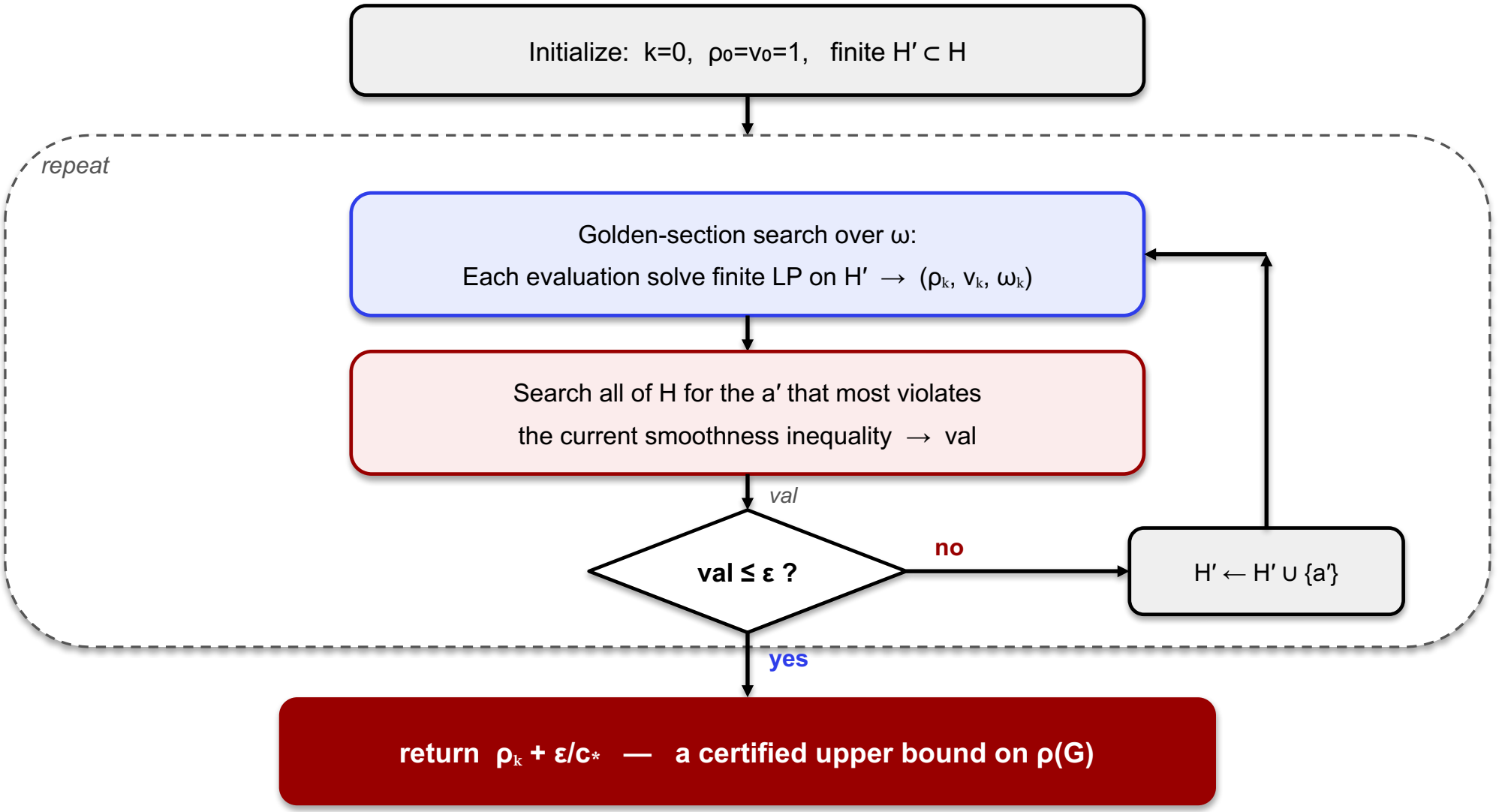
$$\text{s.t.} \quad \Sigma_i\, c_i(\omega\, a_{i,true}, a_{-i}) - \Sigma_i\, c_i(a) + C(a) \leq \lambda\, C(a_{true}) + \mu\, C(a) \quad \forall\ a \in A$$

change of variables →

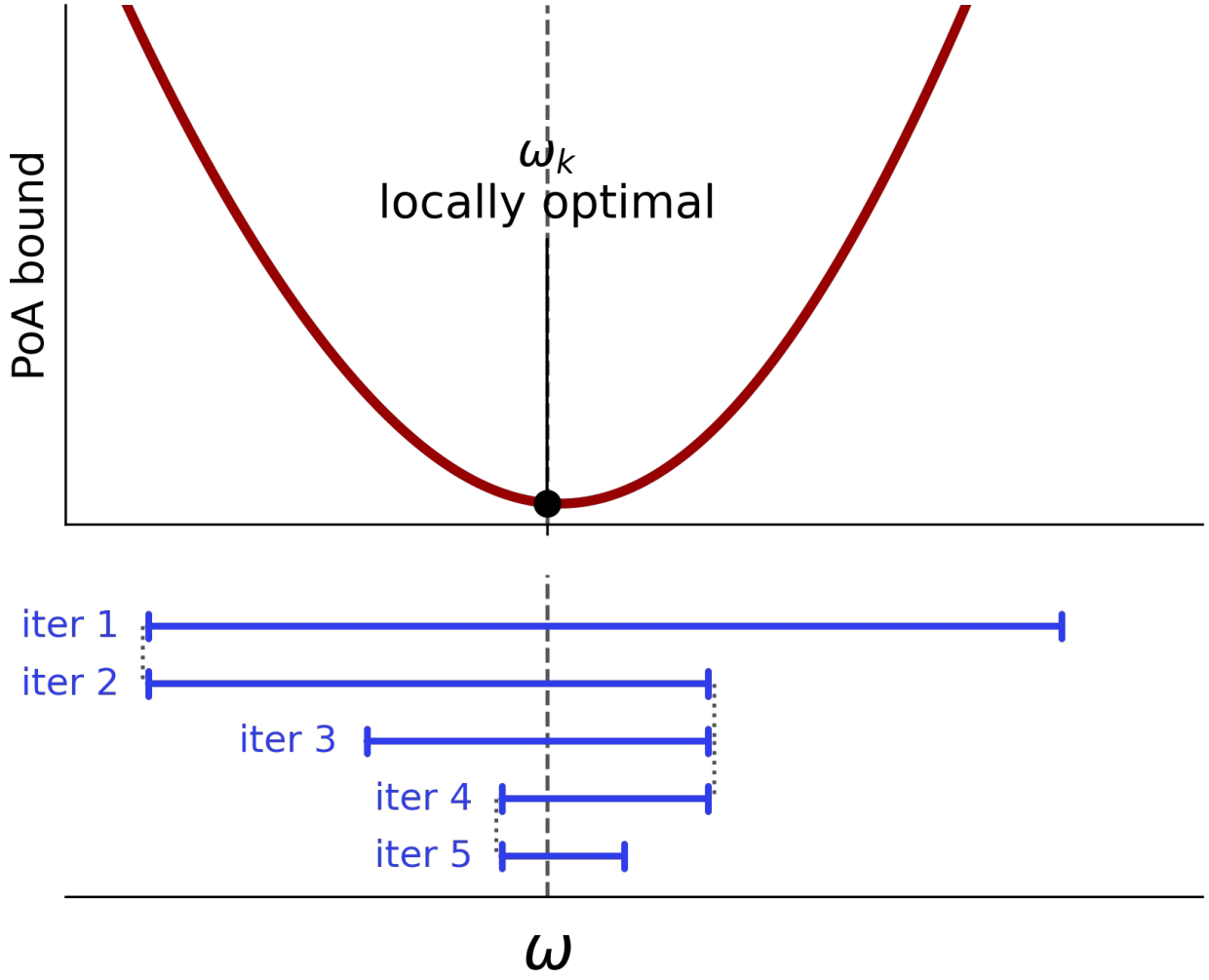
$$\inf_{\rho>0,\ v>0,\ \omega>0} \quad \rho$$

$$\text{s.t.} \quad C(a) + v\,[\,\Sigma_i\, c_i(\omega\, a_{i,true}, a_{-i}) - \Sigma_i\, c_i(a)] \leq \rho\, C(a_{true}) \quad \forall\ a \in A$$

How the bound is computed



Inner loop: golden-section search over ω

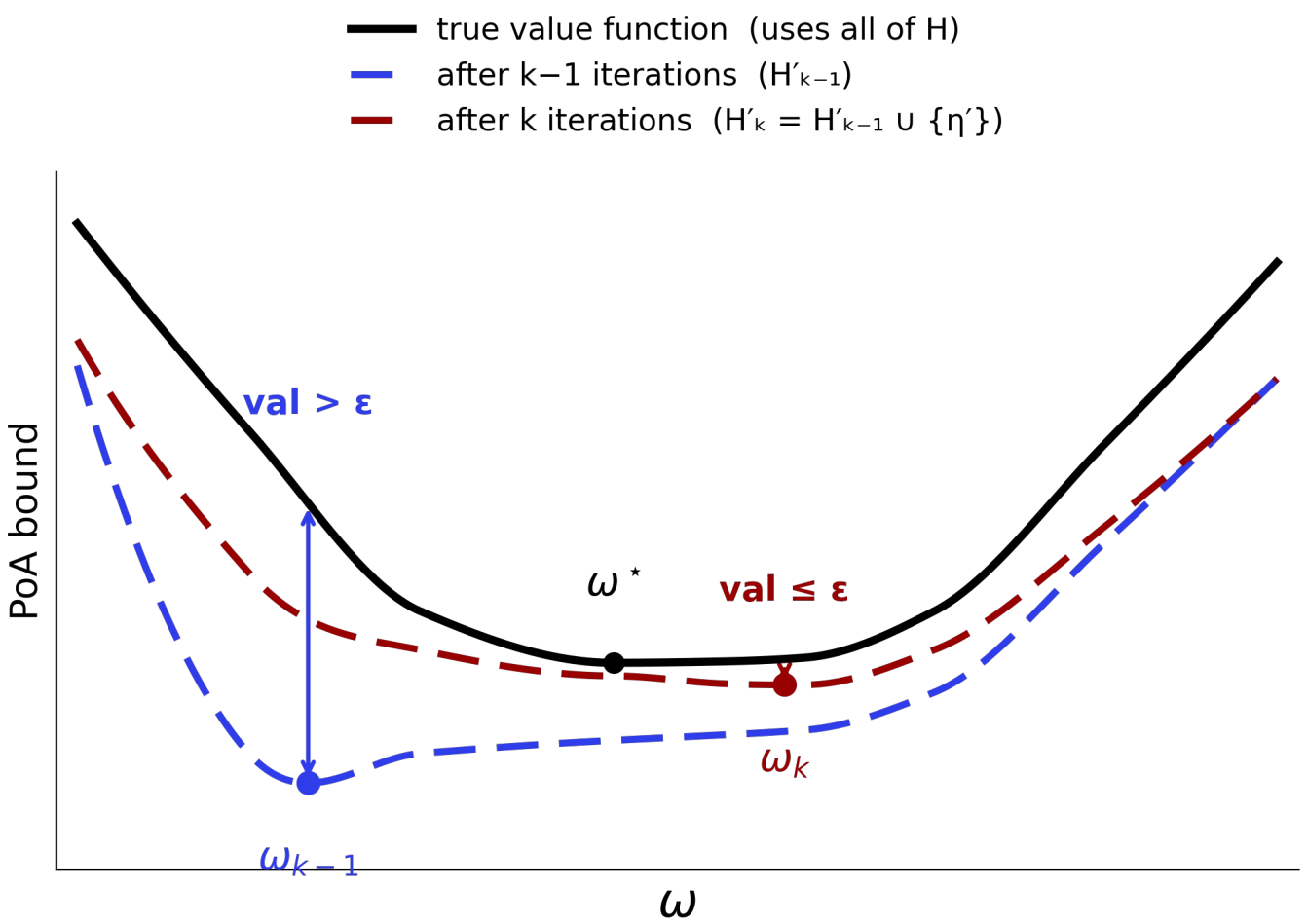


For a fixed ω, the program is linear in (ρ, v), can be solved exactly.

ω is the only nonlinear ingredient, so a 1-D golden-section search over ω finds a local minimum. Each evaluation at a candidate ω just re-solves a finite LP over the current H′.

The solution (ρ_k, v_k, ω_k) goes as candidate into the separation problem of the cutting plane scheme

Outer loop: cutting planes over H



After k−1 iterations, the relaxed value function (using only H'_{k-1}) is optimistic: at its own minimizer ω_{k-1} some smoothness condition is violated by val > ε

The algorithm finds a violating η′ at ω_{k-1} and adds it as a new cutting plane:

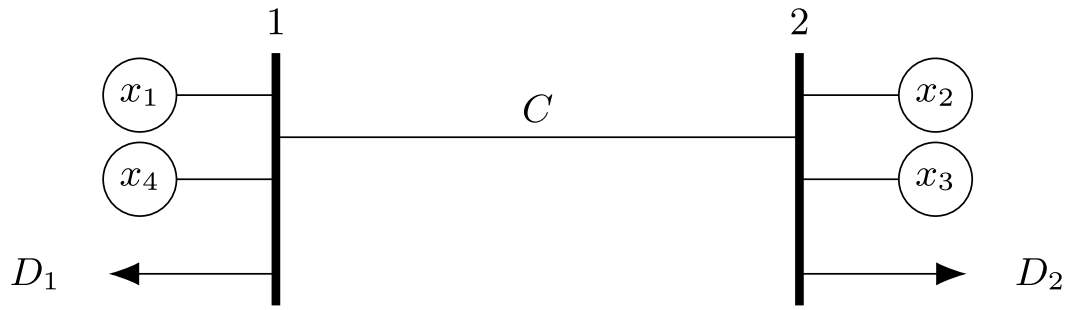
$H'_k = H'_{k-1} \cup \{a'\}$

The new curve need not match the true one everywhere, only near its own minimizer ω_k. If val ≤ ε, $\rho_k + \varepsilon/c_*$ is already a certified bound, even though the two curves still disagree elsewhere.

Numerical results

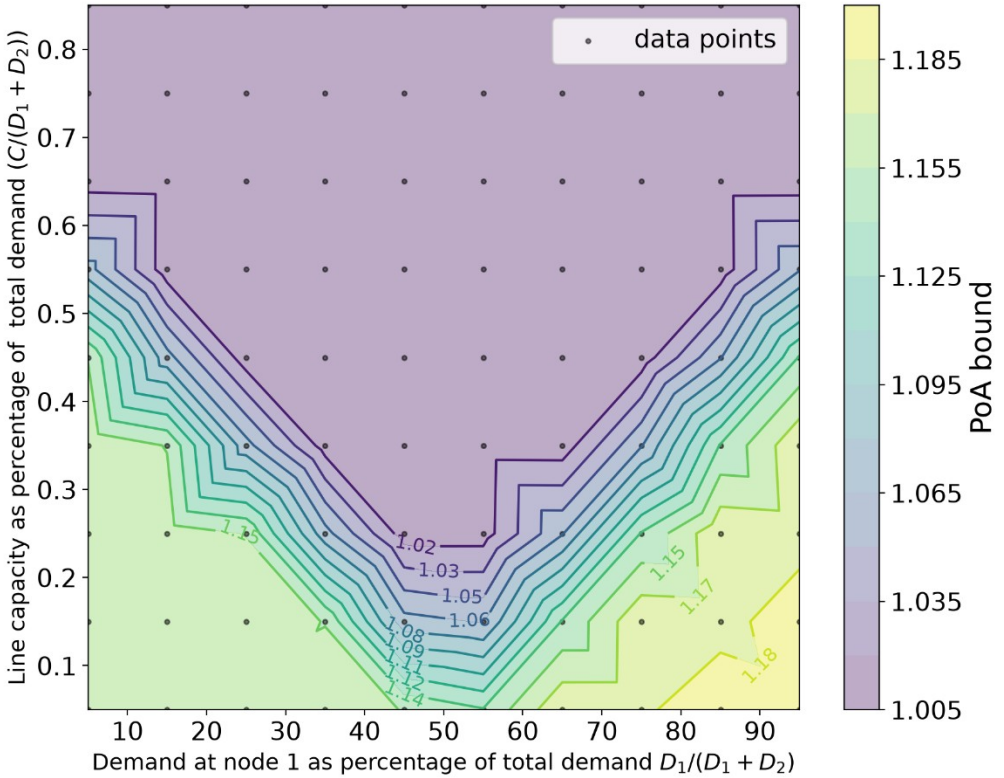
- Considered markets with fully quadratic cost functions
- One time step only
- 4 players per market
- Demand at 1530MW, capacity (if assumed) around 700MW as in DK1 between 9:00-10:00 on August 27, 2023

Sanity in networks



The bound behaves like we would expect PoA to behave:

- Independent of demand ratio for high capacity
- Starts depending on it first for very skewed demand
- Highest PoA when close to two independent markets



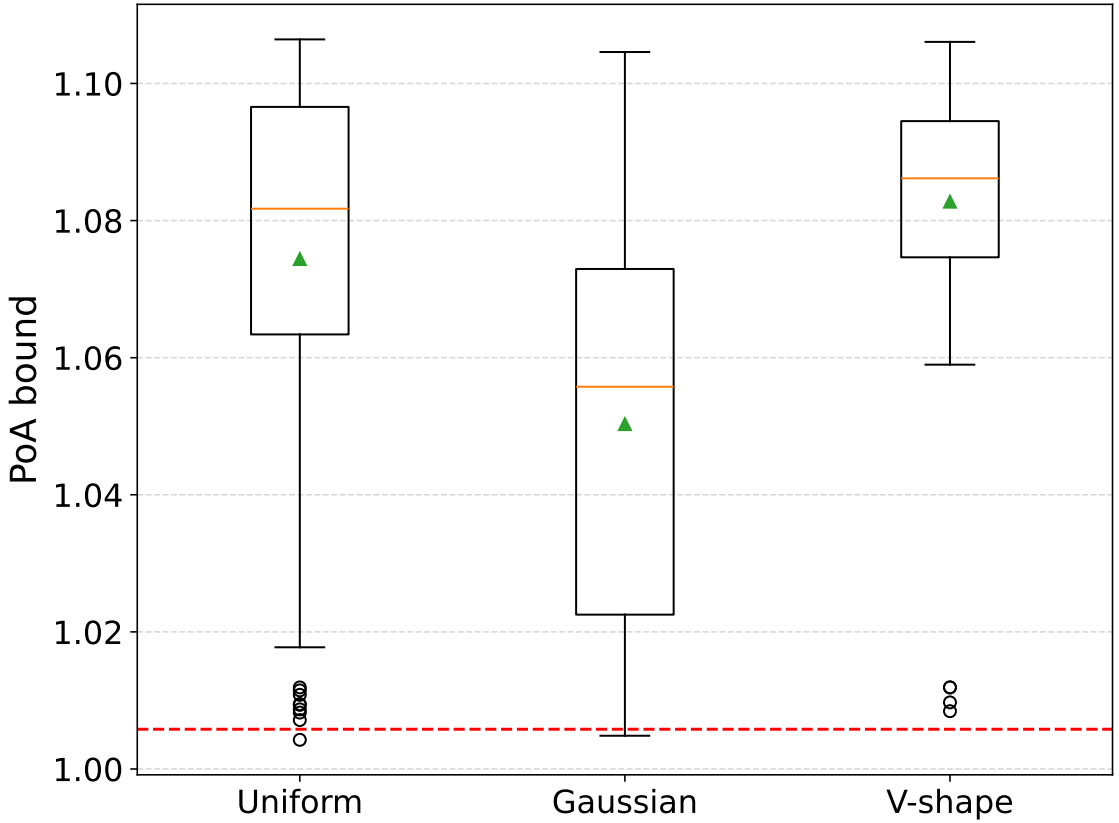
Sanity with capacity constraints

Experiment:

- Consider 100 random markets
- Draw their capacity constraints out of three different distribution with same expected value
- Calculate the bound

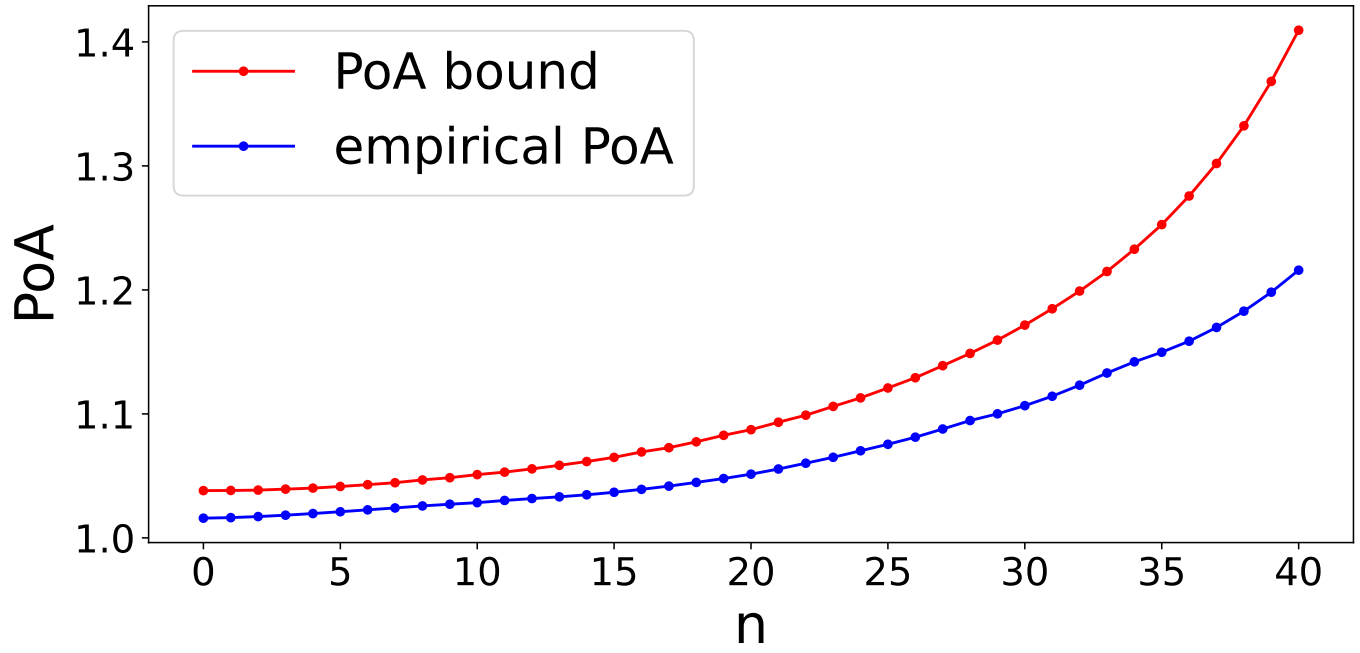
Bound behaves like we would PoA expect to behave:

- Highest when distribution is V-shaped as this induces highest asymmetry between players
- Lowest when distribution is Gaussian as this induces highest symmetry



Quality and sanity - controlled perturbation

Market had true cost parameters 3-0.05n, 3-0.025n, 3+0.025n, 3+0.05n, that is n controls how asymmetric the market is. No physical constraints. Calculated the bound and an empirical value via discretization



We observe:

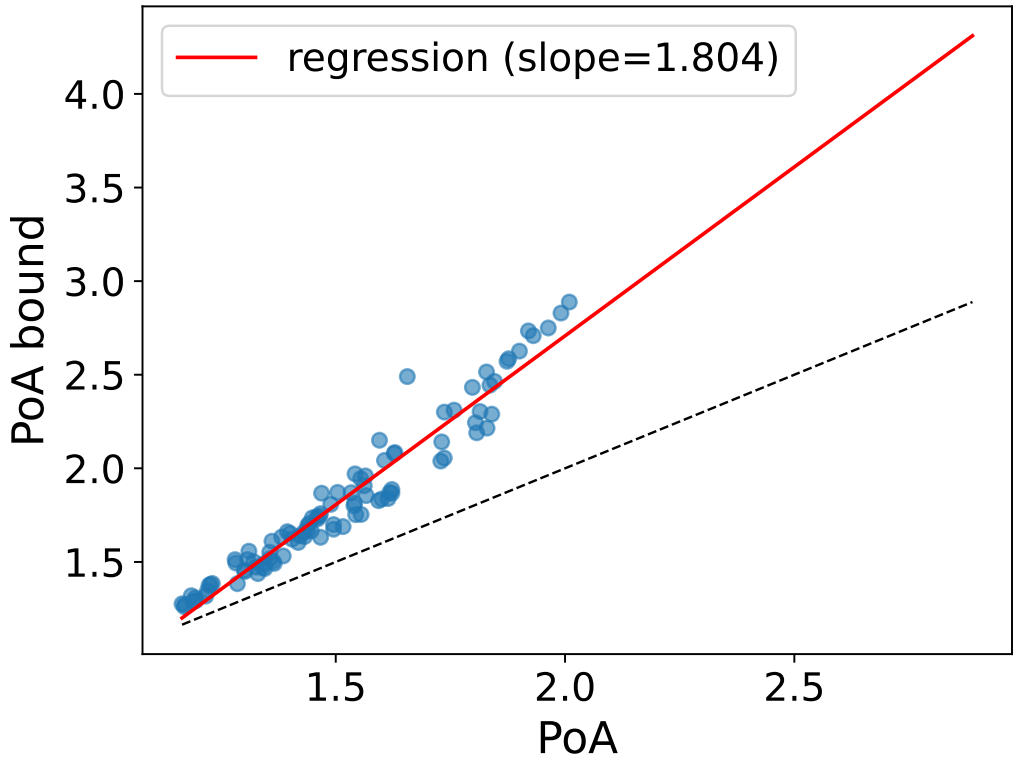
- The bound is especially good when PoA is low
- The bound follows the empirical PoA

Quality and sanity - uncontrolled perturbation

100 random markets without physical constraints. For each calculated bound and empirical PoA via discretization

Observe the following:

- For all markets PoA is smaller than the bound (phew)
- The bound is tighter when PoA is small (as before)
- There is definitely a trend that higher bound means higher PoA but it is not always true



Ideas for projects

- Further sanity-check the bound under physical modeling assumptions: consider many markets with capacity or network constraints and check if the bound follows the prediction on PoA (higher when asymmetry is increased)
- Consider slightly bigger markets and compare bound vs empirical CCE PoA vs empirical NE PoA
- Try to further characterize under which perturbations PoA and PoA bound stay consistent
- Consider what further effect it has on PoA/PoA bound when the bidding language is such that generators can’t even bid their true cost. For example, their true cost is of block form while they can only bid quadratic
- Try to calculate the bound for markets as big and complex as possible (IEEE-systems with several nodes)
- Consider models with multiple time slots, try to decompose the problem to keep it manageable (probably more a thesis topic as it is not clear if that is even possible)

Exercise for this week

- Build groups of ~3
- Discuss which of these ideas you think are interesting; if you have your own idea, I am very open to it!
- After the break we meet to discuss the projects, and you can start immediately
- From then on, no more lectures but only project work (except maybe a lecture that focuses on cool results and colorful examples in the end)


DTU